MD Haffiz Mahamud

PhD Applicant | Computer Vision | Autonomous Perception

mdhafffizmahamud@email.com | linkedin.com/in/md-haffiz-mahamud | Rajshahi, Bangladesh

RESEARCH PROFILE

Mechanical Engineering graduate from Rajshahi University of Engineering & Technology (RUET) working across computer vision, deep learning, sensor fusion, and computational modelling. My research focuses on robust perception in adverse conditions and reliable computational methods. Seeking PhD research opportunities in computer vision, multimodal perception, and autonomous systems.

EDUCATION

B.Sc. in Mechanical Engineering - Rajshahi University of Engineering & Technology (RUET) (2022 - Aug 2026)
Relevant coursework: Numerical Methods, Engineering Mathematics, Control Systems, Fluid Mechanics, Thermodynamics, Heat Transfer, and Engineering Materials.

RESEARCH INTERESTS

Computer vision for autonomous systems; adverse-weather robust perception; multimodal and multi-sensor learning; 3D object detection; LiDAR-radar fusion; uncertainty quantification and confidence calibration.

RESEARCH EXPERIENCE & ONGOING WORK

Severity-Conditioned Object Detection under Fog | **Undergraduate thesis, RUET | 2025-2026**

- Built a fog-aware PyTorch/YOLOv8 detector using learned severity estimation and feature-wise linear modulation to adapt to changing visibility.
- Designed calibration-aware evaluation to examine detection confidence alongside accuracy.

Certified Amortized Homogenization | **Independent research | 2026-present | Manuscript in preparation**

- Developing microstructure-conditioned neural-operator surrogates for conductivity and plane-strain elasticity with rigorous upper and lower effective-property bounds.
- Implemented integration checks, geometry-sensitive bounds, and a bandwidth quality gate; tested synthetic morphologies and segmented sandstone imagery.

LiDAR-Radar Sensor Fusion for Autonomous Perception | **RUET | 2024**

- Designed a dual-sensor distance-estimation pipeline for dusty, foggy, and low-light conditions, comparing Kalman filtering with weighted fusion.

IoT-Based Smart Street Lighting & Embedded Control | **Independent project | 2024**

- Built an Arduino-based adaptive controller integrating ambient-light and motion sensing with real-time hardware testing.

INDUSTRY EXPERIENCE

Industrial Trainee - Rashidpur Gas Field, Sylhet (April 2025)
Observed gas-processing operations, maintenance practices, control systems, and safety procedures.

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, NumPy, SciPy, scikit-learn, pandas, OpenCV; object detection, semantic segmentation, uncertainty quantification and calibration.

Computational Engineering: Neural operators, numerical PDEs, surrogate modelling, PINNs, constrained optimisation, automatic differentiation.

Engineering Tools: SolidWorks, AutoCAD, ANSYS Fluent, MATLAB (basic), finite-element fundamentals.